

1 Input and Output fields

Input and output fields (controls of type mpdvEdit) are used in "LayoutControls" i.e. in the selection panel, in detail applications and in editing dialogs to present data (output) and/or for the entry of data by the user. The controls support a number of features, such as

- Special input controls for the entry of texts, date and/or time values, color selection, true/false values, etc.
- Selection from selection lists (data-based)
- Forwarding to selection dialogs
- Labeling and unit display
- Input of FROM and TO values.

1.1 Customization

You can control all features of an input and output field via customization.

Adopting properties from the MDS Repository

The property *Field name* of an mpdvEdit can reference a property or a service parameter of the MDS repository and this way inherit all properties. In the simplest case, you must only enter a property/service parameter to get a completely defined input and/or output field.

Advantage of this procedure: The relevant properties can be controlled centrally, i.e. all input fields for e.g. machine or personnel numbers have the same behavior and can be changed via central customization settings, if required.

But you can still select any "Field name" for an mpdvEdit and/or you can locally overwrite properties.

Note: If you make a local change, you can no longer control the field centrally.

Manual customization

To manually customize an input and output field, you use the editing mode of a LayoutControl. Enable the MES Development Suite and activate the editing mode using the context menu of the selection panel, for example.. For further information, refer to the paragraph "Selection panel" in section "Customization of applications".

Note the following:

- You can assign controls of the "mpdvEdit" type to a layout using "Add Control". When the editing mode is activated, click a control to show its properties on the right hand side of the dialog "Customization".

- The selection list of the service parameters of the property "Fieldname" is specified via the contents of the "ServiceName" property - only those service parameters are displayed that are known in the service context. If "ServiceName" is empty, all known properties are displayed.



Numerous changes in controls are only shown, when you save, close and restart the application! For example, if you create a new control and assign a field name to it, you must save and restart the application so that the changes become active.

1.2 Properties of Input and Output Fields

You can control the following properties of input and output fields via a property/service parameter or manually via the editing mode:

- **FieldName:** Reference to the property or - if a ServiceName is given - to the service parameter. If a ServiceName is entered, only the parameters of this service are shown, otherwise all known properties are displayed.
- **ServiceName:** Provides the context from which service parameters can be used. The selection list shows all known services.
- **LanguageKey:** Is used to specify the label text and should be a "LanguageKey" ("lkxxx"), so that the text is displayed in the language that has been selected.
- **UnitLabel:** Text key for unit
- **Length:** (in number of characters) controls the field width. If value is 0, the control will use the entire width available to it. Important: If a width is specified that exceeds the available space, the control is cut.
- **ScriptID:** Is used to identify scripts provided by MPDV.
- **ControlType:** Controls the type of controls used. Possible values for ControlType:
 - **TextEdit:** Text field for entry / display of (short) texts. Can be extended by indicating a "ControlDataSource" with a button that opens a search dialog. If you enter the name of a DataLogic in ControlParameter and if a mapping is included in ControlDataSourceResult, data is requested when you leave the control and return values are mapped appropriately.
 - **CheckEdit:** Checkbox for the entry / display of a Boolean value (true/false) or selection of multiple values, if reference to data source is given.
 - **ColorEdit:** Control for entry / display of a color value.
 - **ComboBoxEdit:** Combo box for the input/display of values provided by the data source in "ControlDataSource".

- DateTimeEdit: Control for entry / display of a date and/or a time.
- MemoEdit: Control for entry / display of a deliberate, multi-line text.
- RadioGroup: Control for entry / display of one or more Boolean values (true/false). The values can be controlled using the ControlDataSource.
- ControlTypeMode: Can be used to control the input control. The permitted values depend on the ControlType.
 - CheckEdit: DualState (default), TriState, J;N;J (checked;unchecked;tristate)
 - ColorEdit: none
 - ComboBoxEdit: SingleEdit, Single, Multiple (multiple selection)
 - DateTimeEdit: Date (date display), time (time display), DateTime, RelativeDate (with button for opening the relative date selection, further information: MpdvEdit_RelativeDate)
 - MemoEdit: none .
 - RadioGroup: SingleColumn, SingleRow
 - TextEdit:
 - None; the search button is shown if a ControlDataSource is defined.
 - "SearchButton": Search button is shown.
 - "SearchButtonValidate": Search button is shown. If you enter an invalid value, an error is displayed.
 - "OpenFileDialog" will open a file selection dialog.
- ShowSecondControl: Specifies whether a second control for from/to inputs is shown.
- ControlDataSource: Data source for the selection of values. The data source can be:
 - a Web service. Configuration: ControlType = ComboBoxEdit, ControlDataSourceType = Lookup, ControlDataSource = Entry from tab ControlDataSources, Field Name
 - a RefDat. Configuration: ControlType = ComboBoxEdit (can be RadioGroup), ControlDataSourceType = Reference, ControlDataSource = Entry from tab ReferenceData, column type
 - a pool application. Configuration: ControlType = TextEdit, ControlDataSourceType = Lookup, ControlDataSource = Name of application
 - a script. Configuration: ControlType = ComboBoxEdit, ControlDataSourceType = Script, ControlDataSource = Name of script
- ControlDataSourceMode: mode of data source (Lookup, Reference or Script)

- ControlDataSourceParameter: Request parameter of the data source
- ControlDataSourceResult: Result columns are separated by semicolon. The values are interpreted as follows:
 - First entry: Value
 - Second entry: Labeling
 - Third entry: UnitLabel
 - As from the fourth entry, the fields will be mapped:
 - Via acronym or semantic type.
 - Mapping of fields: in ControlDataSourceResult a mapping in the form "FieldName=SpalteAusResult" can be entered as of the fourth entry. Example: tool.id=resource.id to fill the field "tools.id" with the value "resource.id" from the pool dialog. Several mappings are separated by ";" - spaces are not allowed.
 - Asterisk mapping: Instead of mapping, you can also enter * . Subsequently, all return columns of the pool screen are mapped. The mapping is performed as usual via ID or semantic type.
- ClientDefaultValue: preset default value, see next section 0
- EditableCondition: This value decides whether a field is editable. There are three possibilities:
 - Boolean value: In case of TRUE or FALSE, the field is always editable / non-editable.
 - Binary expression:
 - Field name must be the name of a field that is also located in the ControlPanel.
 - Valid operators: =, <, >, <=, >=, <>, !=
 - The value is written as a string and interpreted depending on the comparative field value.
 - Field, operator and value must be separated by a space!
 - Concatenation of binary expressions
 - You can concatenate an arbitrary number of binary expressions.
 - To link the expressions, you can use &&, AND, ||, OR.
 - Here, too, all components of the conditions must be separated by a space.
 - Evaluation takes place as AND and/or && before OR and/or ||. Parentheses are not allowed.
 - Example: resource.id = 12345 && resource.costcenter = 20 || resource.id = 60610



The default value of the property "ClientDefaultValue" is assigned to the field, if the result of an expression in the EditableCondition or the VisibleCondition changes from FALSE to TRUE. The expressions in the EditableCondition and the VisibleCondition are dynamically evaluated, if the fields of the application change.

- VisibleCondition: Visibility condition. For customization, see EditableCondition.
- LoadDataOnInit: This is used to specify whether the control data is loaded directly when this control is initialized. Some control types within the mpdvEdit contain a list of data (Combobox, LookupCombo, CheckedCombo, RadioGroup) which is filled from RefDat or Lookup values. For reasons of performance, these controls (except for RadioGroup) are always only filled on demand, i.e. when the control is "opened". In some cases, however, it may be reasonable and/or necessary that this takes place upon loading the control already. For this case, the property LoadDataOnInit must be set to true. This, of course, is only reasonable with regard to the controls mentioned. The default value is false.



Not all modifications in the customization files are directly adopted. You must first save and restart the application.

Use of default values in input fields

Input fields have a "ClientDefaultValue" property. The value entered here is displayed as default value when the control is initialized. "From" and "to" values are separated by semicolons.

Set checkbox

If the value for ClientDefaultValue is set to 'true' in a CheckEdit, the checkbox will be set after initializing.

Pre-allocating text fields with "From" and "To" values

You can use a TextEdit to pre-allocate the From and To fields by setting the ClientDefaultValue to 'Value1;Value2'.

Pre-allocation of date fields

With date fields, you can pre-allocate the field with an offset. If you set default values for date fields, you must absolutely specify the type of offset. The following offsets are possible:

- h (hours)
- d (days)
- w (weeks)
- m (months)

- y (years)

The 'to' value is always **relative to the 'from' value**. The default value is always a DateTime object. The presentation depends on the configuration of the relevant field.

You can put "[" and "]" in front and at the end of the relevant value to specify the start and end of a period of time. Consequently, e.g. "[0d;0d]" means that 00:00:00 is entered in the 'from' field today and 23:59:59 is entered in the 'to' field today. "[-1w;0w]" means from Monday last week up to Sunday last week.

Examples

- Current date: 0d
- From today to the day after tomorrow: 0d; 2d
- From today to one week from today: 0d;1w
- From yesterday to tomorrow: -1d;2d
- From one year ago today to one year from today: -1y;2y

Year selection lists

You can use ControlDataSource = YearList and ControlDataSourceMode = Script to create a year selection list or a normal "Service-ControlDataSource". In this case, you can use the following default values:

- Current year: 0y and/or currentyear
- Last year: -1y
- Following year: 1y
- 4 years ago: -4y
- Year that was current 10 months ago: y-10m --> this is usually the case when the relevant year field occurs in combination with a month selection list.

If you want to preallocate two fields (ShowSecondControl), you must separate both values by semicolon, e.g. -1y;1y

Month selection lists

The following default values can be used for a month selection list:

- Current month: 0m
- Last month: -1m
- Following month: 1m

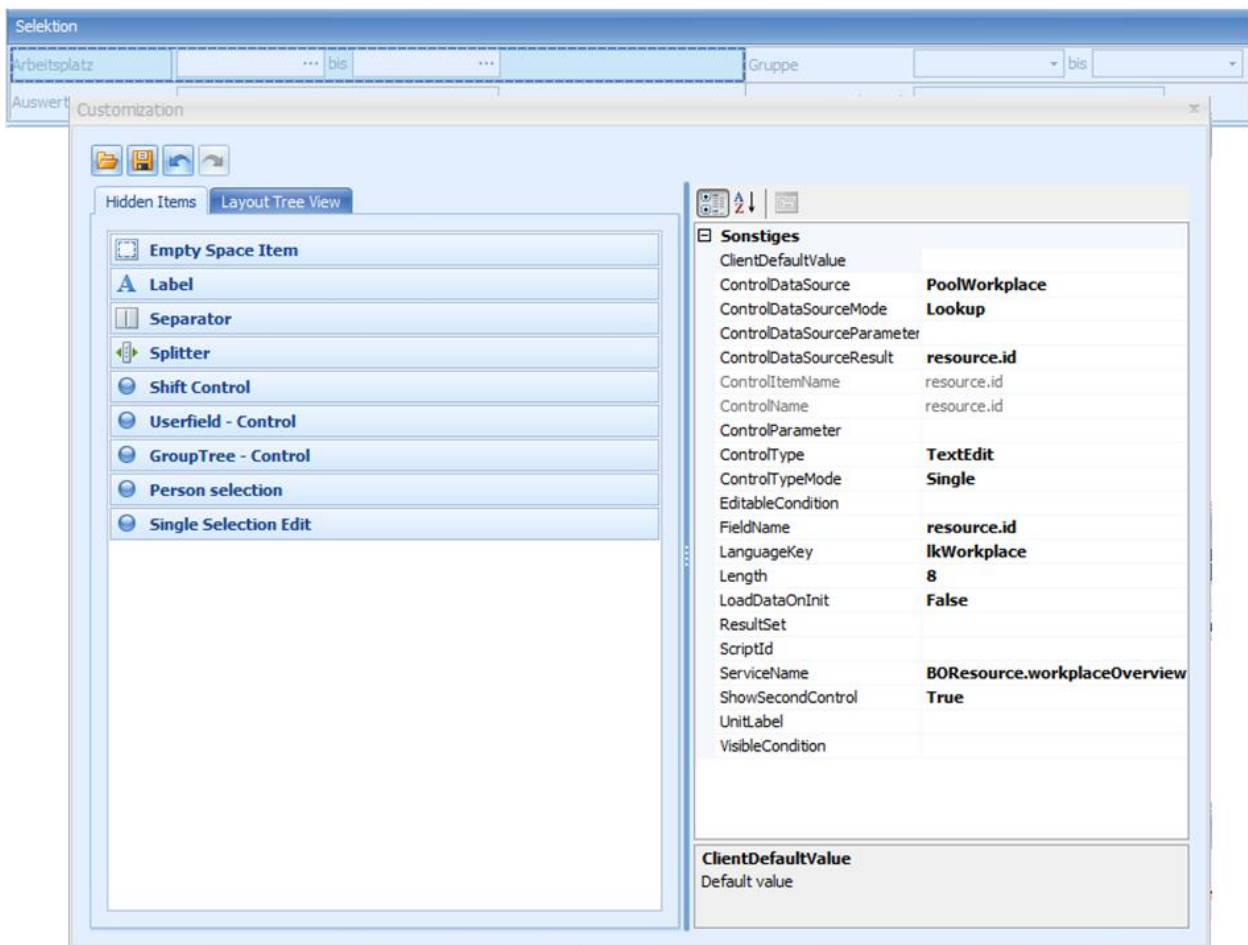
- 4 months ago: -4m

If you want to preallocate two fields (ShowSecondControl), you must separate both values by semicolon, e.g. -1y;1y

1.3 Examples of customization settings

The following examples illustrate possible customization settings of input and output fields.

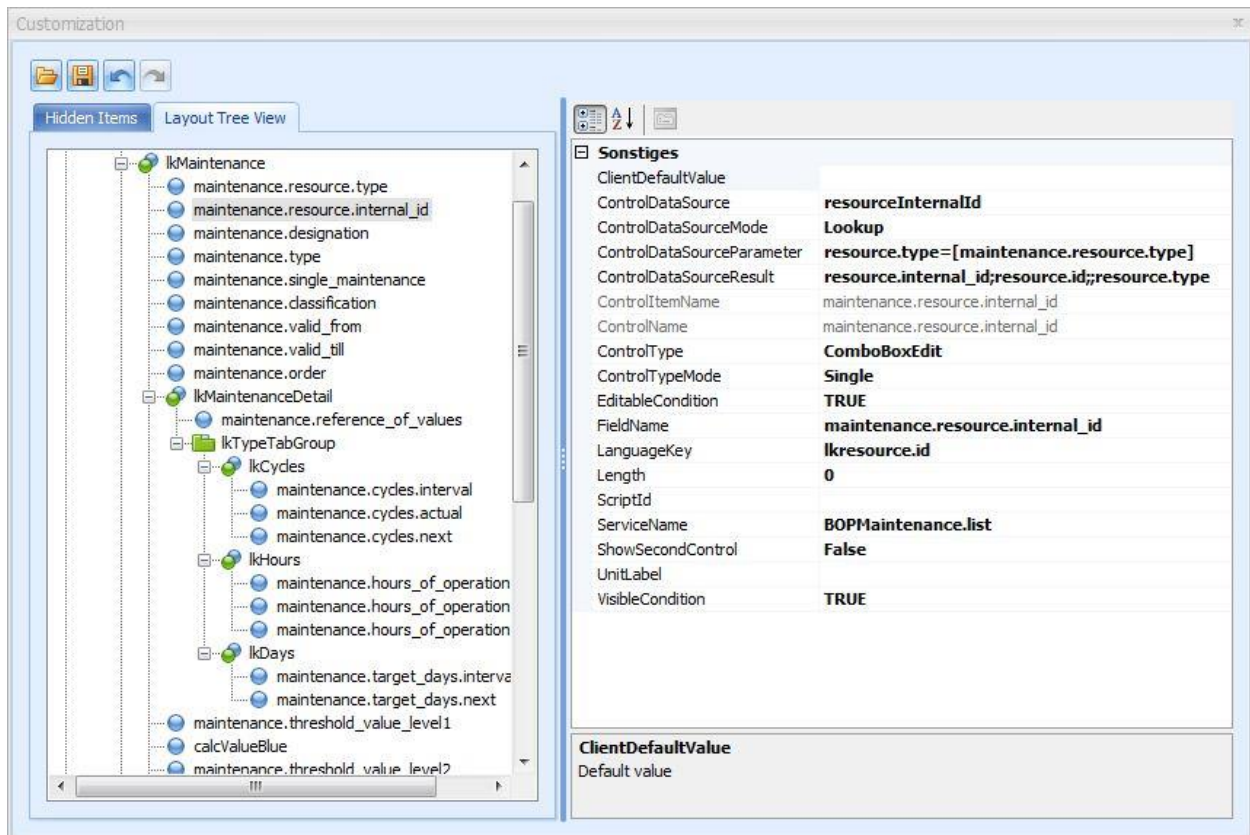
Example: From-To selection parameters with value selection



- ControlType: TextEdit
- ControlDataSource: Application with ID "PoolWorkplace" ("Workplaces" selection)
- ControlDataSourceMode: Lookup
- ControlDataSourceResult: The "resource.id" is selected from the return values of the data record selected in the selection screen and interpreted as result value.
- ShowSecondControl: True => shows the "to" control.

Note: You can only select one value at a time, i.e. a possible "Multiple" ControlType is ignored.

Example: Customization of a selection list



This example shows a complex customization setting for an mpdvEdit. It is an editable ComboBox filled with data from the "resourceInternalId" data source. As filter parameter, the selected value of the "maintenance.resource.type" is transferred to this data source on the same LayoutControl. The data source returns several columns. The column "resource.internal_id" is used as value, "resource.id" is used as display value and "resource.type" is used to fill the "maintenance.resource.type" field.

Example: Data is only requested when ComboBox is opened

Applications with many selection lists (combo boxes) can take longer to open if all combo boxes are filled when the application is requested.

Customization

- ControlDataSource: Name of an existing CDS
- ControlDataSourceMode: Lookup
- ControlType: ComboBoxEdit
- ControlTypeMode: SingleEdit